

PHYSICS PRACTICE EXAMINATION

Comprehensive Review: Mechanics, Energy, and Thermodynamics

Time Allowed: 30 Minutes

Total Questions: 10

1. A ball is thrown vertically upward. At the highest point of its trajectory, which of the following is true?

- A) Its velocity is non-zero and its acceleration is zero.
- B) Its velocity is zero and its acceleration is zero.
- C) Its velocity is zero and its acceleration is 9.8 m/s^2 downward.
- D) Both velocity and acceleration are changing.

2. According to Newton's Second Law of Motion, if the net force acting on an object is doubled while the mass remains constant, the acceleration will:

- A) Be halved.
- B) Remain the same.
- C) Double.
- D) Quadruple.

3. Work is defined as the product of force and displacement in the direction of the force. What are the SI units for work?

- A) Newtons (N)
- B) Watts (W)
- C) Joules (J)
- D) Pascals (Pa)

4. Which of the following best describes the principle of Conservation of Energy?

- A) Energy can be created but not destroyed.
- B) Energy cannot be created or destroyed, only transformed.
- C) The total energy of an open system is always constant.
- D) Kinetic energy is always equal to potential energy.

5. What is the frequency of a wave that has a period of 0.02 seconds?

- A) 20 Hz
- B) 50 Hz
- C) 100 Hz
- D) 5 Hz

6. An object at rest stays at rest, and an object in motion stays in motion unless acted upon by an external force. This is known as:

- A) Newton's First Law
- B) Newton's Third Law
- C) The Law of Universal Gravitation
- D) Ohm's Law

7. If a circuit has a resistance of 10 Ohms and a current of 2 Amperes, what is the voltage across the circuit? ($V = IR$)

- A) 5 Volts
- B) 12 Volts
- C) 20 Volts
- D) 0.2 Volts

8. Which state of matter has a definite volume but no definite shape?

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

9. The heat required to change the state of a substance without changing its temperature is called:

- A) Specific heat capacity
- B) Thermal conductivity
- C) Latent heat
- D) Convection

10. Light travels fastest in which of the following media?

- A) Water
- B) Glass
- C) Air
- D) Vacuum